PERFORMANCE TESTING AND USER ACCEPTANCE TESTING

User Acceptance Testing (UAT) Template

Date	25 October 2026
Team ID	Individual project
Project Name	HotelPro (Hotel Management System)
Maximum Marks	As per evaluation

Project overview

- **Project name:** HotelPro, a web-based hotel management system.
- **Description:** Guests search rooms by date and book and pay online. Front-desk staff manage check-in, check-out and payments. Admins manage rooms and view occupancy and revenue.
- **Project version:** v1.0.0
- **Stack:** React, Node.js, Express, MongoDB, JWT.

Testing scope

- Registration, login, logout and role-based access
- Room search by dates, guests and type
- Booking creation, price calculation and double-booking prevention
- Online payment (simulated), pay at hotel, cancellation and refund status
- Check-in and check-out by staff
- Admin room management and dashboard
- Responsive layout on phone and desktop

Requirements to be tested

- As a guest, I want to find free rooms for my dates and book one easily.
- As a guest, I want to pay securely and cancel an upcoming booking.
- As front-desk staff, I want to check guests in and out and see payment status.
- As an admin, I want to manage rooms and see occupancy and revenue.
- As a user, I want secure login and protected data.

Testing environment

Item	Details
Local URL	http://localhost:5173 (frontend), http://localhost:5000/api/health (API)
Deployed demo URL	_____ (add your deployed link)
Setup	Run the seed script (npm run seed) before testing to load the rooms, demo users and sample bookings.
Admin login	admin@hotelpro.com / Admin@123
Staff login	staff@hotelpro.com / Staff@1234
Guest login	guest@hotelpro.com / Guest@1234
Browsers	Latest Chrome and a phone browser (width 375 px)

Test cases

Expected results are listed for every case. Record the actual result and mark Pass or Fail while executing each case.

Test case ID	Test scenario	Test steps	Expected result	Actual result	Pass / Fail
TC-001	Register a new guest	1. Open Create account. 2. Enter name, email, phone and a password of 8 or more characters. 3. Submit.	Account is created, user is logged in and sent to the rooms page. Navbar shows the user name.		
TC-002	Register with a short password	1. Open Create account. 2. Enter a password of 5 characters. 3. Submit.	Error says the password must be at least 8 characters. No account is created.		
TC-003	Register with an existing email	1. Register using guest@hotelpro.com.	Error says the email already exists.		
TC-004	Log in with valid details	1. Open Log in. 2. Enter guest@hotelpro.com and Guest@1234. 3. Submit.	Login succeeds. My bookings link appears in the navbar.		
TC-005	Log in with a wrong password	1. Enter a valid email and a wrong password. 2. Submit.	Error: Incorrect email or password. No session is created.		
TC-006	Protected page while logged out	1. Log out. 2. Open /bookings directly.	User is redirected to the login page.		
TC-007	Guest cannot open the dashboard	1. Log in as guest. 2. Open /admin directly.	User is redirected to the home page. No dashboard data is shown.		
TC-008	Search rooms by dates	1. On the home page choose check-in tomorrow and check-out in 2 days. 2. Set 2 guests and type Double. 3. Search.	Only Double rooms that sleep 2 or more and are free on those nights are listed.		
TC-009	Search with one date only	1. Choose a check-in date and leave check-out empty. 2. Search.	Message asks for both dates or neither. No search is run.		
TC-010	Book a room	1. Log in as a guest. 2. Open a room. 3. Choose dates and guests. 4. Click Confirm booking.	Booking is created. Total equals nights x nightly rate. It appears in My bookings as confirmed with payment pending.		
TC-011	Guests above room capacity	1. Try to book with more guests than the room sleeps.	Booking is rejected with a message showing the room capacity.		
TC-012	Prevent double booking	1. Book a room for certain dates. 2. With another account, try to book the same room for overlapping dates.	Second booking is rejected with a clear message. The room no longer appears in a search for those dates.		
TC-013	Pay online	1. Open My bookings. 2. Click Pay online. 3. Choose UPI and pay.	Payment status becomes paid. A transaction reference and method are shown.		
TC-014	Cancel an upcoming booking	1. Click Cancel booking on a confirmed booking. 2. Confirm.	Status becomes cancelled. The room appears again in searches for those dates.		
TC-015	Cancel a paid booking	1. Pay for a booking. 2. Cancel it.	Booking is cancelled and payment status becomes refunded.		
TC-016	Staff views and filters bookings	1. Log in as staff. 2. Open Dashboard, then Bookings. 3. Filter by each status.	All bookings are listed. The filter shows only the chosen status.		
TC-017	Check in a guest	1. Find a confirmed booking whose arrival date is today. 2. Click Check in.	Status becomes checked-in.		
TC-018	Check in before the arrival date	1. Find a confirmed booking arriving in the future. 2. Click Check in.	Action is rejected: check-in is only possible from the arrival date.		

Test case ID	Test scenario	Test steps	Expected result	Actual result	Pass / Fail
TC-019	Check out a guest	1. Find a checked-in booking. 2. Click Check out.	Status becomes checked-out.		
TC-020	Record a pay-at-hotel payment	1. Find a booking with payment pending. 2. Click Mark paid.	Payment status becomes paid with a cash reference.		
TC-021	Dashboard figures	1. Log in as admin. 2. Open Dashboard, Overview.	Guests in house, arrivals, departures due, occupancy, rooms and revenue match the bookings shown.		
TC-022	Add a room	1. Log in as admin. 2. Open Rooms and click Add room. 3. Fill the form and save.	Room appears in the admin list and on the public rooms page.		
TC-023	Close a room for maintenance	1. Edit a room and set status to Maintenance. 2. Save. 3. Search as a guest.	Room no longer appears in the public search.		
TC-024	Delete a room with active bookings	1. Try to delete a room that has a confirmed booking.	Deletion is blocked with a message. The room stays in the list.		
TC-025	Guest list	1. Log in as staff or admin. 2. Open the Guests tab.	Registered guests are listed with email, phone, booking count and last stay.		
TC-026	Responsive layout	1. Open the site at 375 px width. 2. Visit rooms, booking, My bookings and login.	Menu collapses into a toggle. Pages are usable with no sideways scrolling.		
TC-027	Log out	1. Click Log out. 2. Open /bookings.	Navbar shows Log in. The bookings page redirects to login.		

Performance testing

Targets below come from the non-functional requirements. Record the measured result after testing on the deployed demo.

Metric	Target	How to measure	Measured result
Home page load time	Under 2 seconds	Chrome DevTools, Network tab or Lighthouse	
Room search API response	Under 500 ms	DevTools Network tab (time for the /api/rooms request)	
Booking creation API response	Under 500 ms	DevTools Network tab (POST /api/bookings)	
Login API response	Under 500 ms	DevTools Network tab (password hashing adds some time)	
Concurrent users	50 simultaneous searches with no errors	Load tool such as autocannon or k6 against /api/rooms	

Bug tracking

Bug ID	Bug description	Steps to reproduce	Severity	Status	Additional feedback
BG-001	First request is slow after the free-tier backend has been idle.	Open the demo after about 15 minutes of no use.	Low	Known (hosting plan)	First load can take about 30 seconds. Open the demo once before a review, or use a paid plan.
BG-002	Two guests sending the same booking at the same instant could both pass the availability check.	Send two identical booking requests at exactly the same time.	Low	Known limitation	Availability is checked before saving. A database transaction or unique reservation key would remove the gap.

Sign-off

Tester name	
Testing period	
Date	
Signature	

Notes

- Cover both positive and negative scenarios for every feature.
- Record detailed feedback and suggestions for improvement.
- Note severity, status and steps to reproduce for every bug.